

SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

ELECTRONIC INSTRUMENTATION
MODEL QUESTION PAPER

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all questions in one word or one sentence. Each question carries one mark.

(9 x 1 = 9 Marks)

1	Define the accuracy of an instrument.	M1.02	U
2	List two types of errors	M1.02	R
3	Write the equation for the torque in permanent magnet moving coil (PMMC) mechanism.	M2.01	U
4	Write one difference between analog and electronic multimeter.	M2.02	U
5	Name a bridge that can be used for inductance value measurement	M3.01	U
6	Name one AC bridge.	M3.01	R
7	What is the use of Horizontal deflection plates in CRT	M4.01	U
8	Name a recorder that can be used for recording the VI characteristics of a diode	M4.04	U
9	Write any one comparison of LED and LCD.	M4.03	U

PART B

II. Answer any eight questions from the following. Each question carries 3 marks

(8 x 3 = 24 Marks)

1	Explain the difference between accuracy and precision	M1.02	U
2	Explain the sensitivity of an instrument	M1.02	U
3	Illustrate the conversion of PMMC mechanism into ammeter.	M2.02	U
4	Explain the difference between moving coil instruments and moving iron instruments	M2.01	U

5	Derive the expression for unknown capacitance in a Scherings bridge	M3.01	U
6	Compare between AC and DC bridges	M3.01	U
7	Explain the working of LCD	M4.03	U
8	Draw the Wein's bridge circuit and write the equation for frequency	M3.01	U
9	Illustrate the difference between gross error and systematic error	M1.02	U
10	Illustrate the advantages of Digital Storage Oscilloscope (DSO) over Cathode Ray Oscilloscope (CRO)	M4.02	U

PART C

Answer all questions. Each question carries seven marks

(6 x 7 = 42 Marks)

III	Explain the generalized block diagram of instrumentation system.	M1.01	U
	OR		
IV	Illustrate any three Dynamic characteristics of an instrument.	M1.02	U
V	Explain working of Analog multimeter with block diagram.	M2.02	U
	OR		
VI	Explain working of Digital multimeter with block diagram.	M2.03	U
VII	Derive the balance equation of an AC bridge.	M3.01	U
	OR		
VIII	Explain working of induction type single phase energy meter	M3.02	U
IX	Explain working of DSO with block diagram.	M4.02	U
	OR		
X	Illustrate the working of XY recorders, with necessary figures.	M4.04	U
XI	Illustrate any two static characteristics of an instrument.	M1.02	U
	OR		
XII	Illustrate any three types of errors in an instrument.	M1.02	U

XIII	Explain working of CRO with block diagram.	M4.01	U
	OR		
XIV	With necessary figures illustrate the working of strip chart recorders.	M4.04	U

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PART A

I. Answer all questions in one word or one sentence. Each question carries one mark.

(9 x 1 = 9 Marks)

1	Define the term precision.	M1.02	U
2	List any four dynamic characteristics.	M1.02	R
3	Write the equation for the torque in moving iron instruments	M2.01	U
4	Write one advantage of digital multimeter over analog multimeter.	M2.02	U
5	Write the equation for frequency in Wein bridge.	M3.01	R
6	Name any two bridge circuit used for inductance measurement	M3.01	R
7	What is the use of control grid in CRO?	M4.01	U
8	Name a recorder that can be used to plot the variation of any physical quantity with time.	M4.04	U
9	Write any one advantage of LED over LCD.	M4.03	U

PART B

II. Answer any eight questions from the following. Each question carries 3 marks

(8 x 3 = 24 Marks)

1	Explain the difference between repeatability and reproducibility	M1.02	U
2	Explain hysteresis with reference to an instrument.	M1.02	U
3	Illustrate the conversion of PMMC mechanism into ammeter.	M2.02	U
4	Explain the differences between analog and electronic multimeter.	M2.01	U
5	Derive the expression for unknown capacitance in a Schering bridge	M3.01	A

6	Explain the construction of dynamometer type watt meter.	M3.02	U
7	Explain the working of LCD	M4.03	U
8	Illustrate the advantages of strip chart recorder.	M4.04	U
9	Illustrate any two types of errors in instruments.	M1.02	U
10	Write three differences between AC and DC bridges.	M3.01	U

PART C

Answer all questions. Each question carries seven marks

(6 x 7 = 42 Marks)

III	Explain any six static characteristics of an instrument.	M1.02	U
	OR		
IV	Explain the main functional elements of a measurement system.	M1.01	U
V	Illustrate the construction and working of moving coil instruments.	M2.01	U
	OR		
VI	Explain working of Digital multimeter with block diagram.	M2.03	U
VII	Derive the balance equation of a Wheatstone bridge.	M3.01	A
	OR		
VIII	Explain the construction and working of single-phase energy meter.	M3.02	U
IX	Explain working of CRO with block diagram.	M4.01	U
	OR		
X	Illustrate the working of X-Y recorders, with necessary figures.	M4.04	U
XI	Explain different systems of units and their conversions.	M1.01	U
	OR		
XII	Explain gross error, systematic error and random error with examples	M1.02	U

XIII	Explain working of seven segment LED with diagram.	M4.03	U
OR			
XIV	With necessary figures illustrate the working of potentiometric recorders.	M4.04	U